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Technology of Metallic Materials A.Y 2025-26

Mechanical Engineering

Steels: Strengthening by Heat Treatment

(Tempered steels)



Giovanni Maizza
DISAT

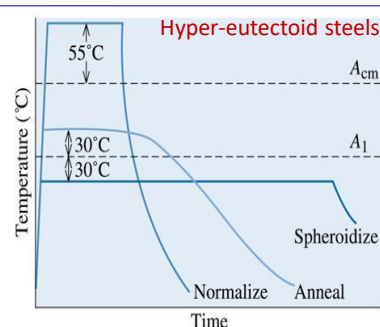
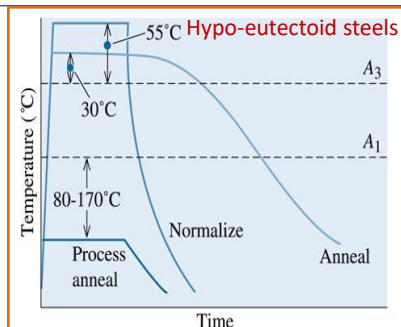
Lecture 9

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Conventional Heat Treatment of hypo- and hyper-eutectoid steels



Austenitization: holding in the austenitic range at $A_3 + 50^\circ\text{C}$ (max solubility, max chemical homogeneity) while preventing grain growth

Annealing: high temperature diffusion at 30°C above A_3 (max chemical homogenization) to achieve max softening of an equilibrium microstructure

Normalizing: austenitization + air cooling (moderate refinement) to achieve good mechanical properties at low cost (slightly non equilibrium)

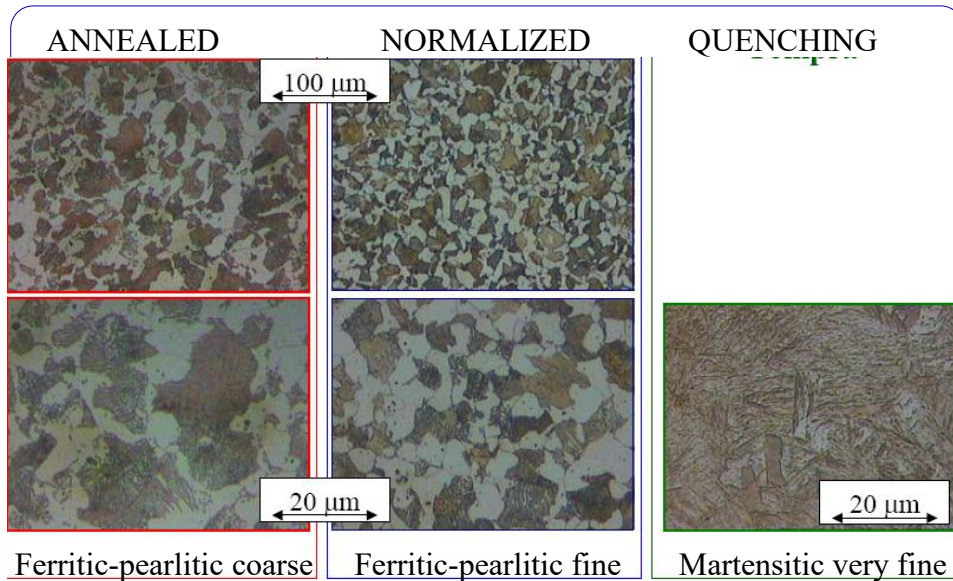
Spheroidizing: pendulum heating at A_1 (spheroidization of pearlite and cementite) to achieve the softest ever (tough) condition (for hyper-eutectoid steel)

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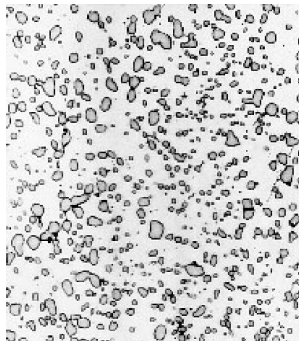
Example of Microstructure after Heat Treatment: AISI1045



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Spheroidizing Annealing



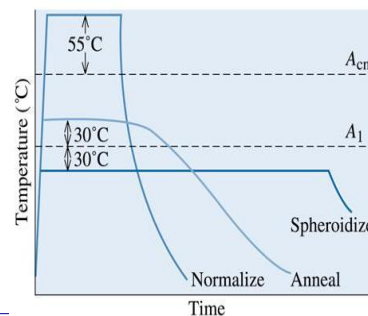
Spheroidized steel; the small particles are Iron carbides; the grey matrix is ferrite 1000x.
Copyright 1971 - United States Steel Co.

This microstructure in hyper-eutectoid steels promotes the **largest ductility** which useful in shaping operations

Prolonged annealing to T just below A_1

Pearlite or Bainite $\xrightarrow{\text{heating}}$ spheroidal cementite
18 – 24 h
 $\approx$ just below A_1

Decrease in interfacial area $\propto \theta$



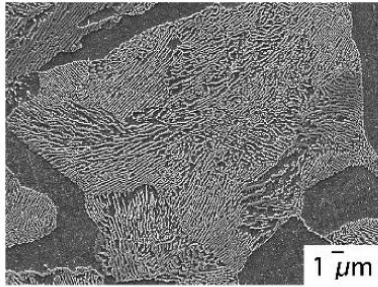
(b) Hypereutectoid

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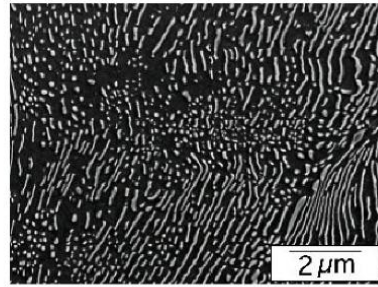


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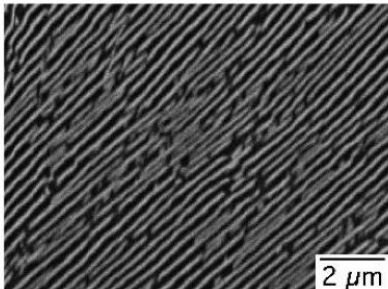
Subsequent steps during spheroidizing



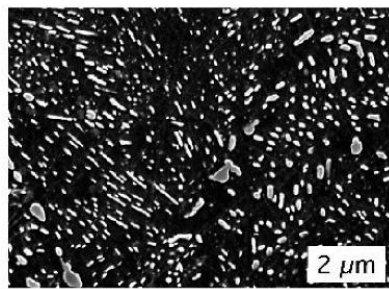
Pearlite with 0.24 micrometers interlamellar spacing



After spheroidisation for 1.5 h at 750 centigrade



Pearlite with 0.13 micrometers interlamellar spacing



After spheroidisation for 1.5 h at 750 centigrade

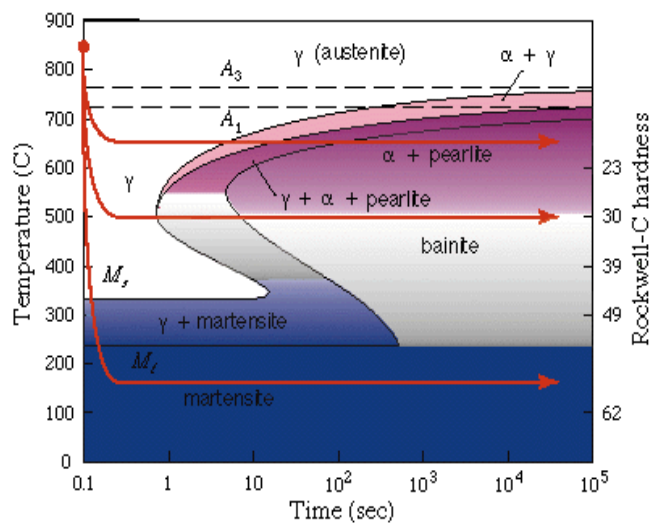
Ferrer et al., 2005

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ISOTHERMAL H.T.



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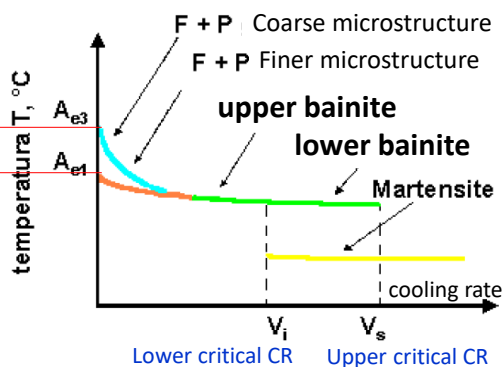
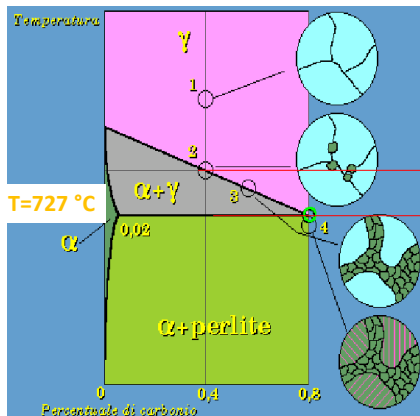
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Phase Transformation on Continuous Cooling

Cooling rate (CR) rather than undercooling (ΔT) is the controlling factor for phase transformation
Large CR operates as Large ΔT and small CR operates as small ΔT

Example: equilibrium cooling steel C=0.4%)

Example: continuous cooling steel C=0.4%)

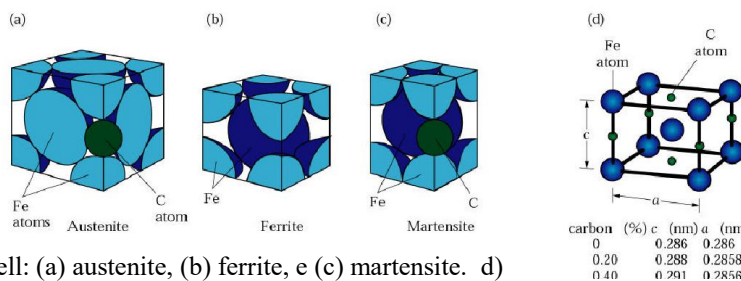


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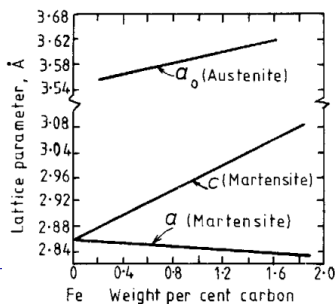


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MARTENSITE STRUCTURE



Unit cell: (a) austenite, (b) ferrite, c (c) martensite. d) effect of %C (%wt) on lattice constant. Note the interstitial position of C atoms and the consequent increase of c lattice constant with increasing %C; these effects cause a lattice distortion of the U.C. of martensite which, from BCC ferrite becomes tetragonal structure.

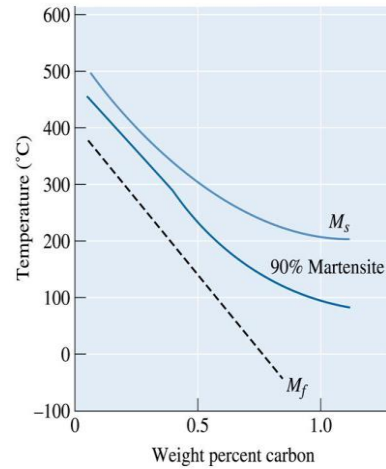
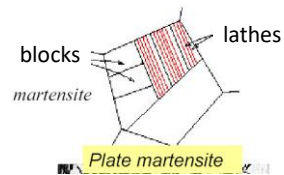
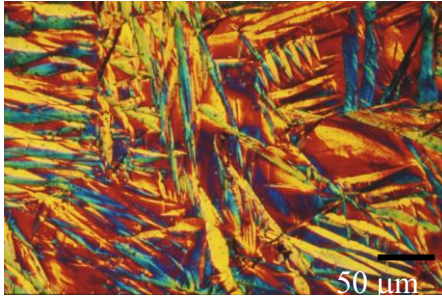


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MARTENSITE STRUCTURE

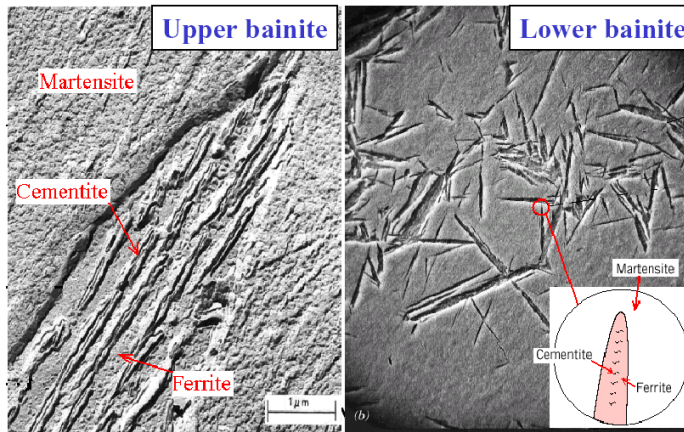


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Upper and Lower Bainite (500 °C – M_s)



“rods”

“acicular”

Optimum tradeoff between mechanical strength and toughness.

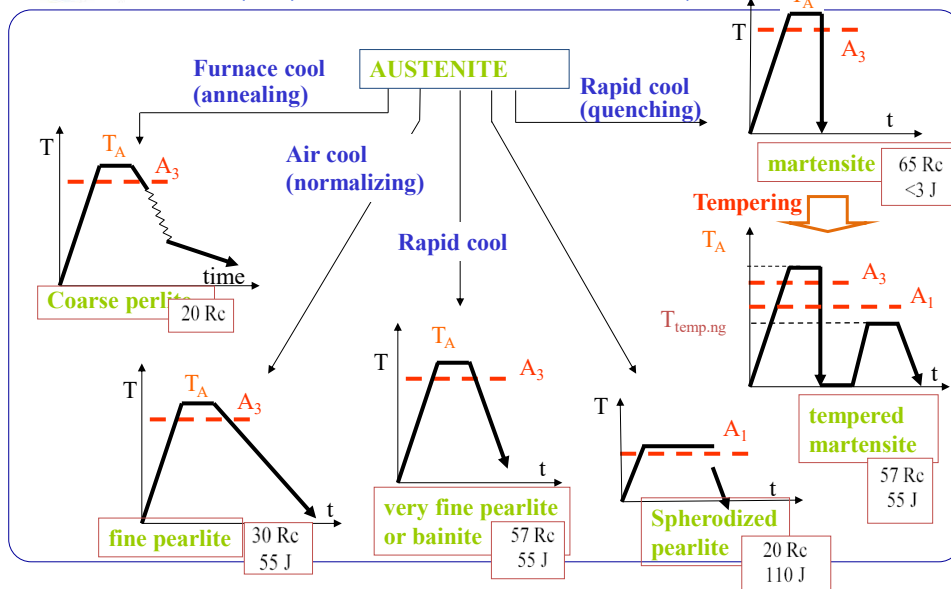
Excellent alternative to tempered martensite for hardness (tempered steels).

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Microstructure, Rockwell Hardness, CVN, Cooling rate (CR) in various Heat Treatment (H.T.)



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QUENCH SEVERITY OF COOLING MEDIA

Medium	QUENCH SEVERITY	
	H Coefficient	Cooling Rate at the Center of a 1-in. Bar (°C/s)
Oil (no agitation)	0.25	18
Oil (agitation)	1.0	45
H ₂ O (no agitation)	1.0	45
H ₂ O (agitation)	4.0	190
Brine (no agitation)	2.0	90
Brine (agitation)	5.0	230

CR changes with the penetration depth inside the component

Hot component immersed in a quenching bath

$Q = H A (T - T_{fl})$
 heat extracted by the quenchant

fluid (quenchant)

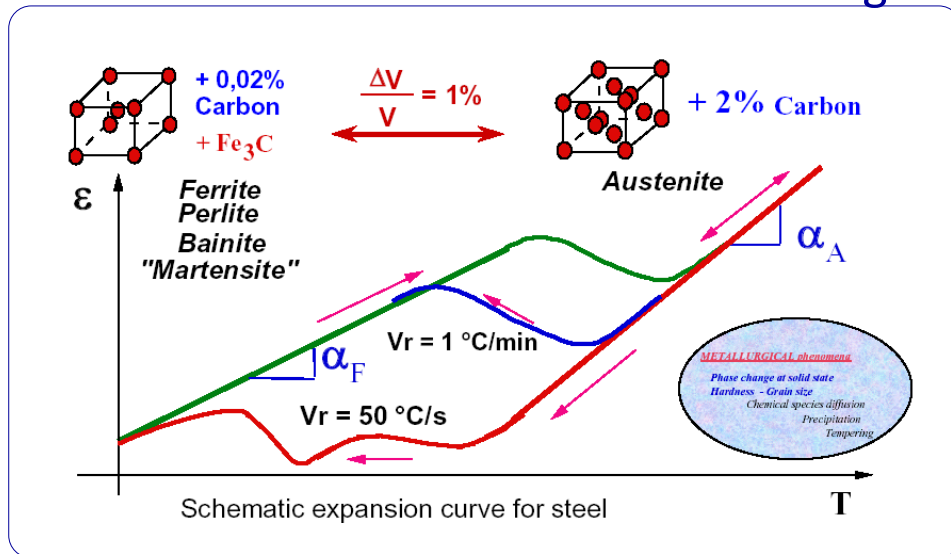
T_{amb}

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Volume changes in phase transformations on cooling

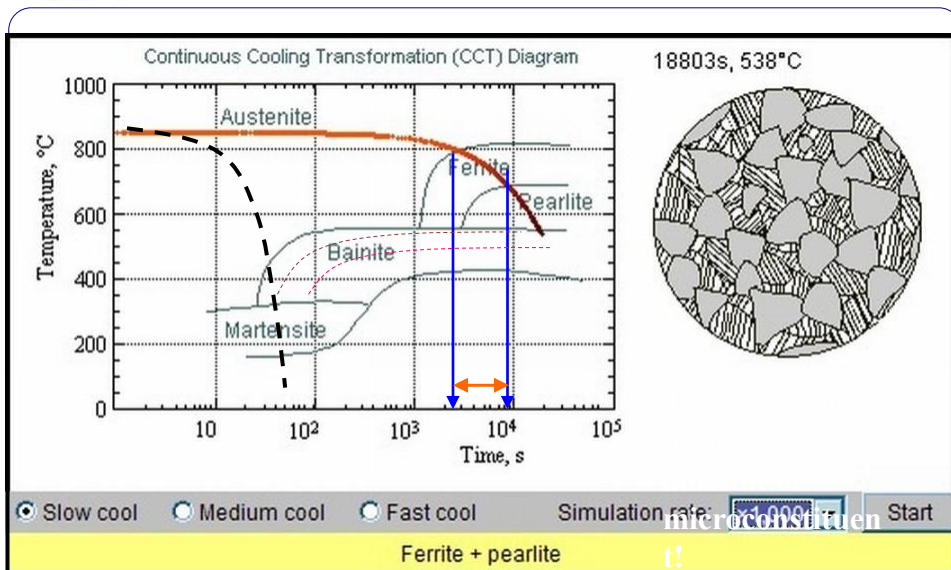


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CCT curves: slow cooling

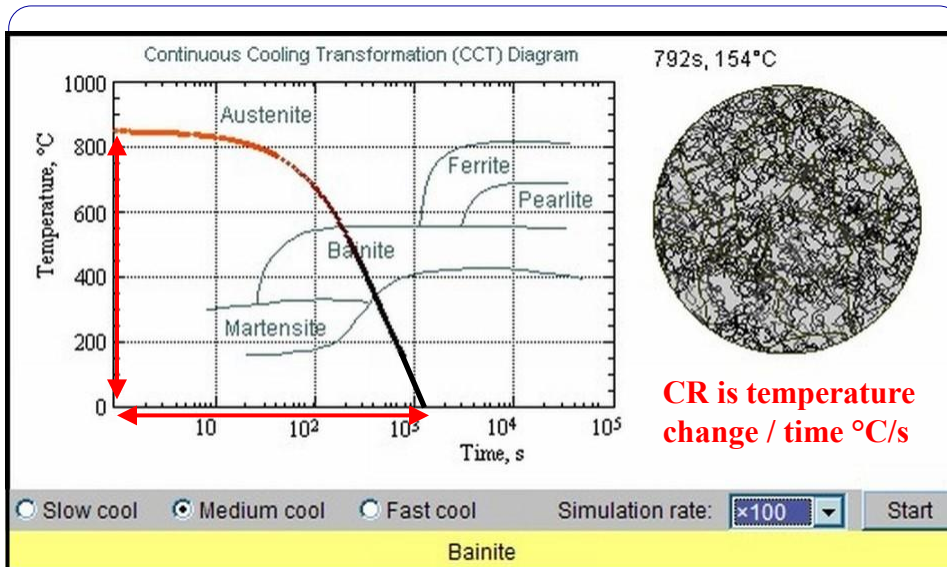


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CCT curves: moderate cooling

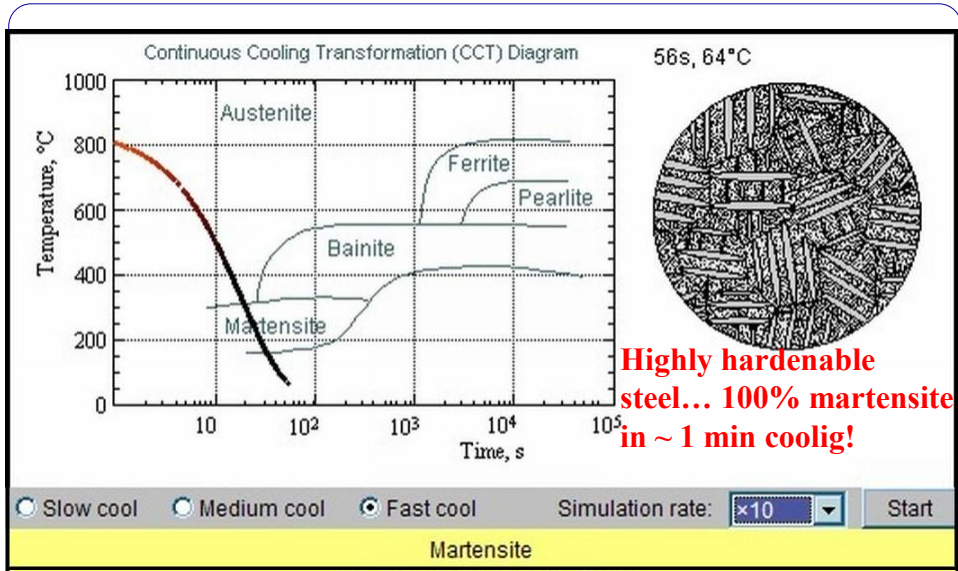


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CCT curves on Quenching

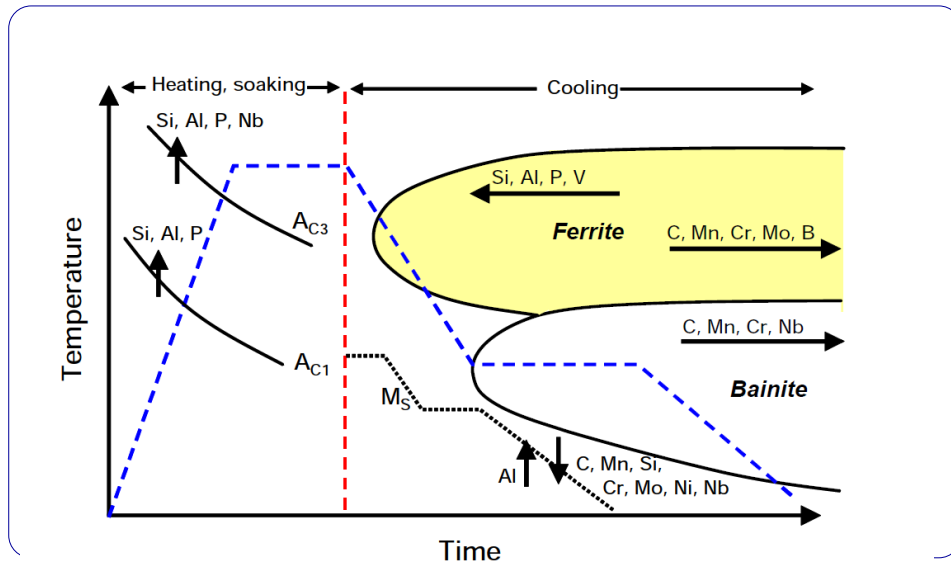


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Influence of alloying elements on transformation behaviour during Continuous Annealing of TRIP steels (as-cold rolled)



Bleck ebook

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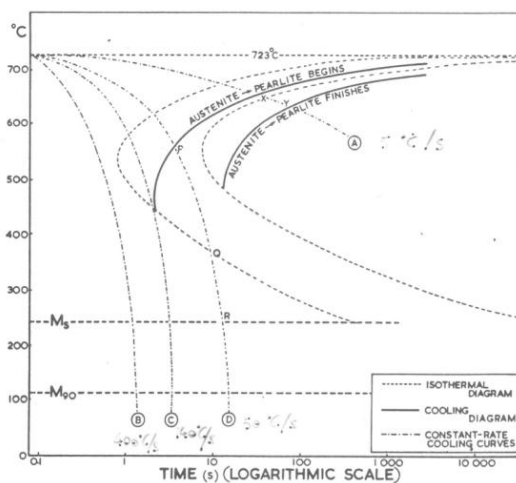


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CCT CURVES vs TTT CURVES

CCT curves are shifted
bottom-right in comparison
to TTT curves.

Incubation times in CCT
are longer than in TTT
curves



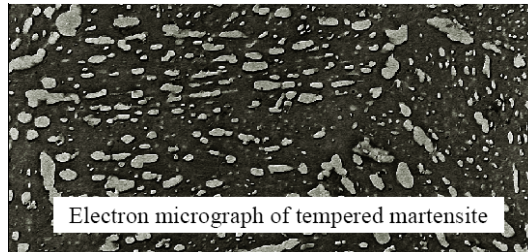
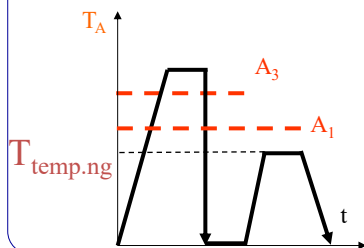
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Microstructure of Tempered Martensite

- Martensite is extremely hard and brittle (**quenching determines very severe residual stresses which may even crack the part**)
- Tempering is done in the range 250 to 650 °C
- After tempering, martensite is an extremely fine mixture of ferrite and Fe_3C precipitated from the quenched martensite having a
- more spheroidal shape thus imparting a
- marked increased toughness to the final tempered martensite.



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Mechanical properties of Tempered carbon steels

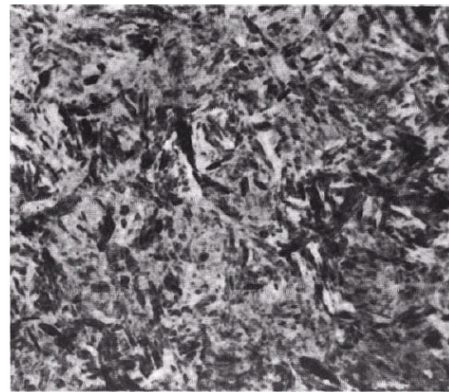
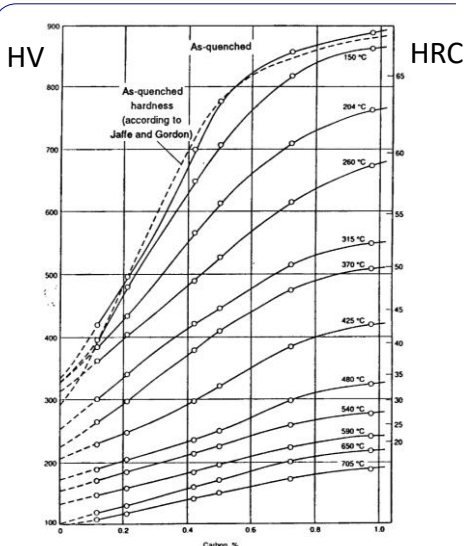


Figure 10-18 The microstructure of tempered martensite, although an equilibrium mixture of α -Fe and Fe_3C , differs from those for pearlite (Figure 9-2) and bainite (Figure 10-9), 825 $\times$. This particular microstructure is for a 0.50 wt % C steel comparable with that described for Figure 10-16. (From Metals Handbook, 8th Ed., Vol. 7: Atlas of Microstructures, American Society for Metals, Metals Park, Ohio, 1972.)

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Alloying Effects in Tempered Steels

- **C**: $\uparrow$ (Y_s , HRC) but $\downarrow$ CVN (toughness), $\uparrow$ hardenability (H); combines with (Cr,Ti,Ta,W,V,Mo)
- **Mn**: its role depends on %C; with $\uparrow$ Mn, $\downarrow$ ductility and weldability. Mn has strong effect on hardenability ($\uparrow \uparrow$ H); strong austenitizer after Ni
- **Al**: powerful **deoxidizer** in the melt, grain refiner, detrimental when $>0.1\%$
- **Si**: **deoxidizer**; less effective than Mn in $\uparrow$ HRC. When excessively $\uparrow$, it is detrimental for surface finish; $\uparrow$ elastic limit
- **Cu**: when excessively $\uparrow$ it is detrimental in hot worked steels as $\downarrow \downarrow$ surface finish. If $>0.20\%$ (but $<0.5\%$) $\uparrow$ Rcorr
- **Ni** ferrite strengthener: $\uparrow$ UTS while maintaining high ($\uparrow \uparrow$ CVN, $\Delta L\%$); preferentially present in solid solution; strong austenitizer
- **Mo** $\uparrow$ Rcreep at high T; $\uparrow \uparrow$ H, overcomes the temper -embrittlement problem;
- **B**: $\uparrow \uparrow \uparrow$ H
- **Cr, Ti, Ta, W, Nb, V, Si, Mo**: strong carbide former and grain refiners, $\uparrow \uparrow$ HRC without depressing toughness; $\uparrow$ resistance to high T; $\uparrow$ resistance to wear

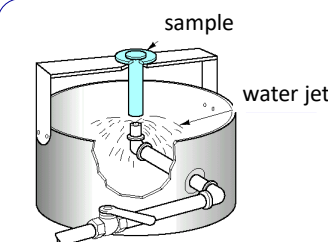
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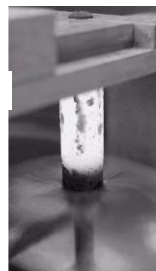


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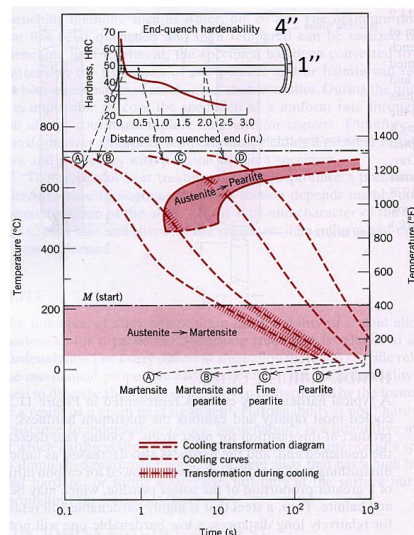
Hardenability (H): a material property



The Jominy end quench test



- **Hardenability**: the ability of the steel to form at least 50% martensite in depth.
- Measured by the Jominy test
- Mo, Mn and Cr largely influence H.

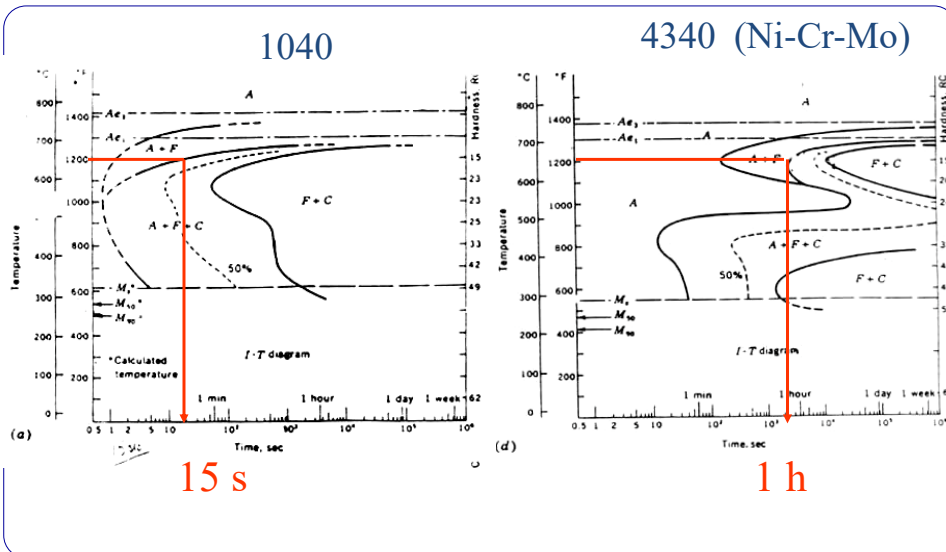


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TTT curves: hardenability

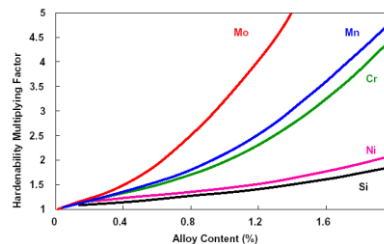
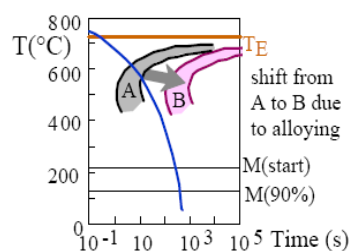


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Hardenability (H): martensite penetration



- Jominy end quench results, $C = 0.4\text{wt}\%C$

Tempered low alloy steels

4140 : 1.0 Cr / 0.2Mo

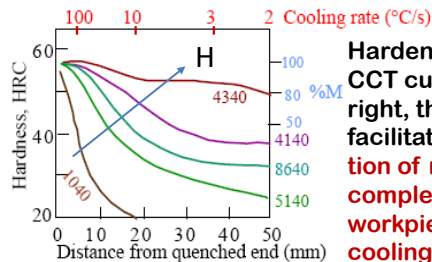
4340: 1.85Ni / 0.8Cr/0.25Mo

5140: 0.85Cr

8640: 0.55 Ni/ 0.5Cr/ 0.20Mo

these elements shifts the nose;

- martensite is easier to form



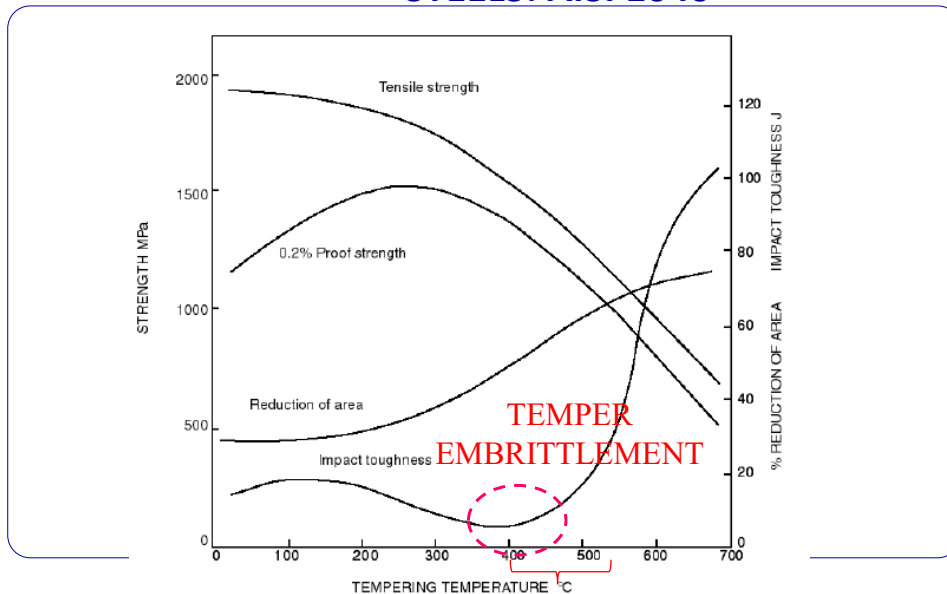
Hardenability shift the CCT curves towards right, thereby facilitating the **formation of martensite in complex or large workpieces with low cooling rate**

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EMBRITTLEMENT OF TEMPERING STEELS: AISI 1040



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PROBLEMS AFTER TEMPERING: Temper-embrittlement (or Krupp's illness)

EVENT: a quenched steel, whenever tempered at increasing T ($> 300^\circ\text{C}$), its K increases but it experiences a dramatic drop when:

- the intercritical temperature range is crossed (ΔT_{IC}) of $400\text{--}550^\circ\text{C}$ (on heating);
- after tempering at $T > \Delta T_{IC}$, the steel is slowly cooled across ΔT_{IC} ;
- although cooling rapidly up to the temperature inside the critical interval it is held for a certain time at such T and then cooled rapidly up to T_{amb}

The loss of toughness increases with the longer holding time within the critical temperature range.

Condition: reversible (recoverable)

Fracture: intergranular

Remedy: +Mo (suff. 0.2%); or +W (0.4%); tempering at $T < \Delta T_{IC}$; if after tempering at $T > 600^\circ\text{C}$ it is slowly cooled in the critical temperature range, the steel is recoverable if heated again at $T > 600^\circ\text{C}$ rapidly cooled across ΔT_{IC}

This problem does not exist in plain carbon steels ($\text{Mn} < 0.7\%$) but only in low alloy steels, with $\uparrow$ Mn and Cr (e.g. low alloy steels with Mn, or with Cr, Cr-Mn, Cr-Ni); it is also present in grain-coarsened steels, since it increases the probability for segregations; absent or negligible in steels with Cr-Mo, Cr-Ni-Mo

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Possible Explanation for Krupp's illness

:

- Precipitation of impurities (P, As, Sb, Sn) with some alloy elements (Mn e Cr), in submicroscopic size around grain boundaries

From Cr- and Mn- carbides parts of Mn and/or Cr will dissociate in the critical interval, by combining with impurities.

In contrast Mo and W will form more stable complex carbides which are stable with Cr and Mn, hence will retard the precipitation thus avoiding embrittlement.

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Inconveniences on tempering: residual austenite

On quenching: the transformation $\gamma \rightarrow m$ may be incomplete, it occurs when $T_{\text{quench}} > M_f$. Note that $M_s = f(\% \text{elem dissolved into the matrix}) \Rightarrow$ with $\uparrow \%C$ and other austenitizing alloy elem., $\uparrow \% \gamma_{\text{res}}$; with $\uparrow CR$, $\uparrow \% \gamma_{\text{res}}$; on **stepped quenching**, $\uparrow \% \gamma_{\text{res}}$, because the isothermal soaking stabilizes γ

On tempering: the γ content that can be transformed at room temperature depends on $\% \text{alloy elem}$: the higher the content of alloy elements, the higher the tempering temperature, the larger fraction of residual γ .

Consequences of residual γ : $\downarrow$ HRC, Rfatigue, $\uparrow$ dimensional instability; internal stresses (γ transforms very slowly and determines volume changes).

Goal: the intent is not to completely eliminate γ , rather to impede the formation of metastable γ in the finished product; austenite imparts resilience, sometimes it can be useful; the only problem is that sooner or later it will transform into martensite or bainite.

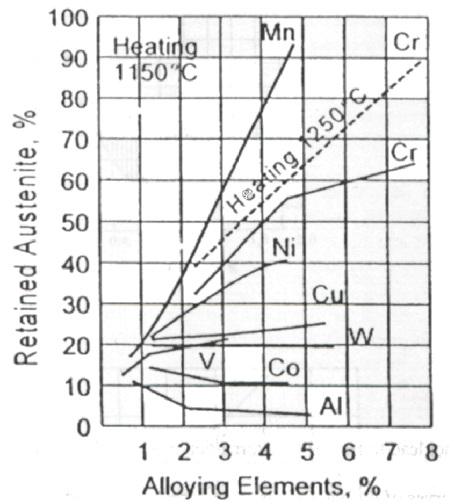
Solution: execute **two or more tempering treatments** (e.g. carburizing steels); **quenching under-zero degree C** (e.g. -80°C with dry ice and acetone; up to -140 and -150°C , in liquid nitrogen) in order to surpass M_f + **relaxation** (e.g. medium C alloy steels);

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Residual Austenite vs %element additions



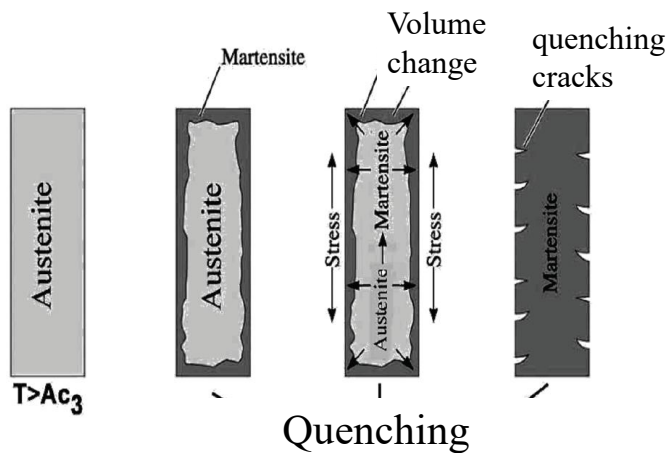
Steel Heat Treatment handbook, p.61

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RESIDUAL STRESSES AFTER QUENCHING



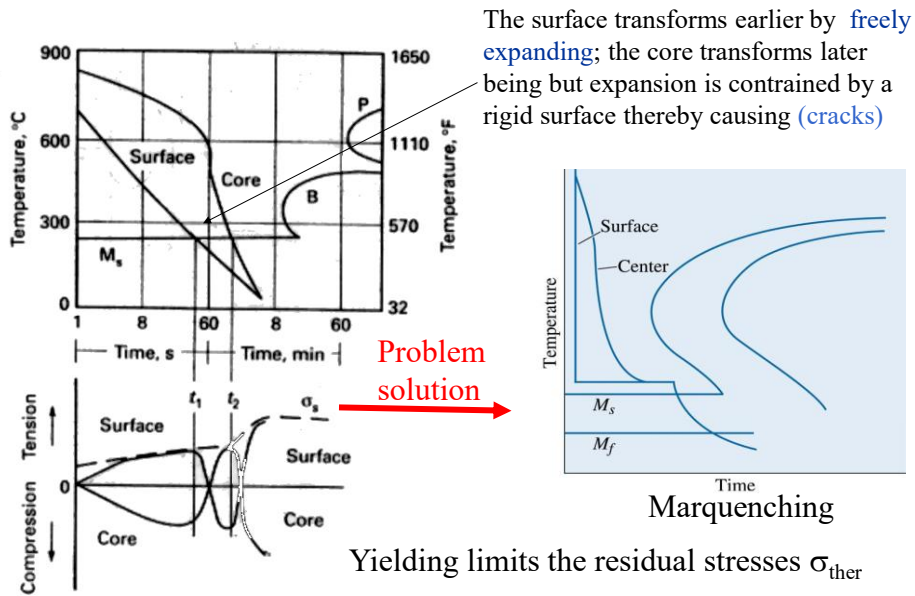
The austenite-martensite transformation on quenching causes an increase in volume resulting in quenching cracks. This is frequent in large pieces and complex geometry parts

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Prevention of Quenching Cracks



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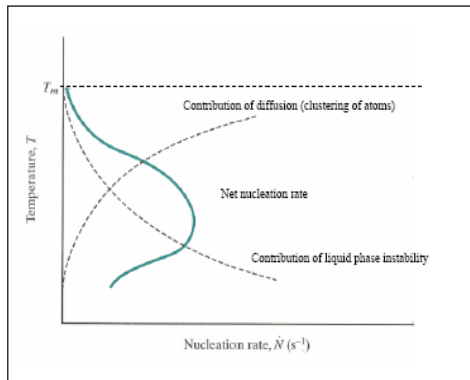
END

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Nucleation rate



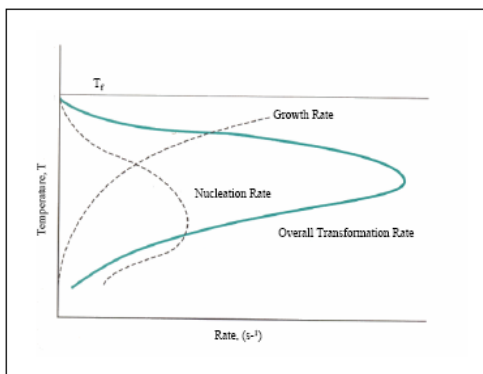
- The overall nucleation rate is a product of two opposite working factors:
 - the driving force for solidification **increases** with T decrease;
 - while the diffusion that is responsible for the clustering process **decrease** with T decrease.
- These explain the “knee-shaped” nature of the transformation curve with maximum at some value below T_f

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Growth rate of a new phase



- The overall transformation rate is a product of two factors:
 - nucleation**;
 - growth**, which is thermally activated process, typically described by Arrhenius function

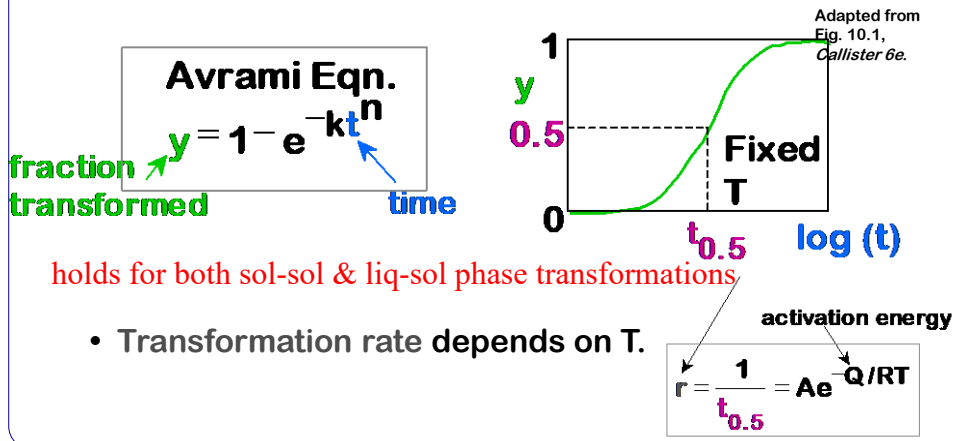
PT rate achieves **maximum** at some T below T_f !!

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ISOTHERMAL TRANSFORMATION GROWTH KINETICS

- **volume (fraction) transformed** depends on time.

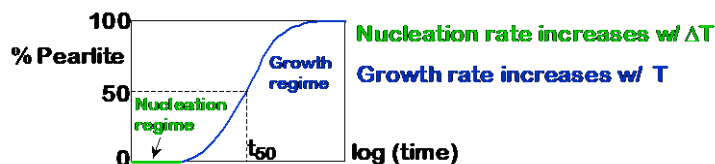


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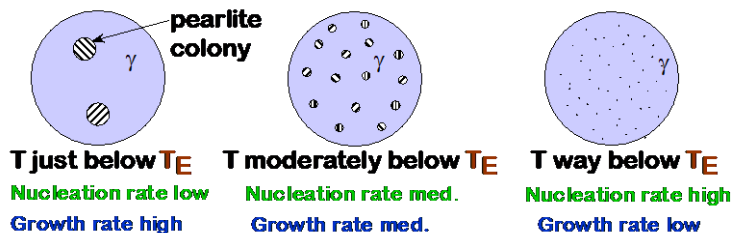
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ISOTHERMAL PHASE CHANGE ex. Pearlite formation

- Reaction rate is a result of nucleation and growth of crystals.



- Examples:



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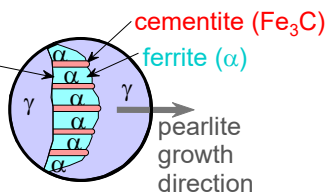


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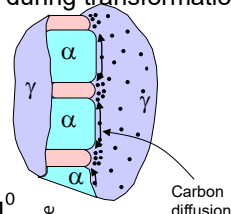
ex. Pearlite Growth on slow cooling

- Transformation of austenite to pearlite:

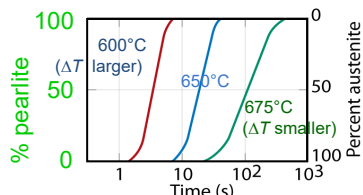
Austenite (γ)
grain boundary



Diffusion of C
during transformation



- For this transformation, rate increases with (ΔT) $[T_{\text{eutectoid}} - T]$.



Coarse pearlite → formed at **higher temperatures** – relatively soft

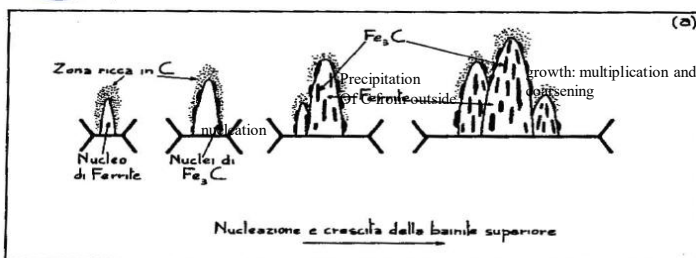
Fine pearlite → formed at **lower temperatures** – relatively harder

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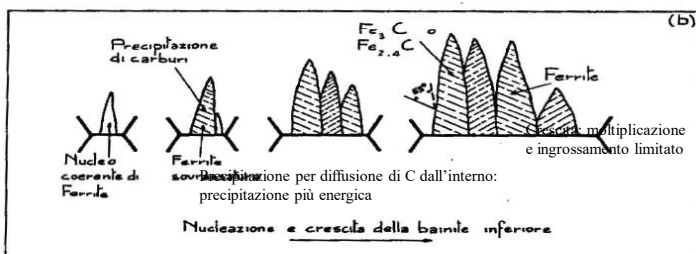
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BAINITE FORMATION: Acicular: ferrite+ carbide needles



Isolated rods in ferritic matrix

Equilibrium structures are still present



Isolated rods mixed with needle-like carbides ($\alpha + \theta$)

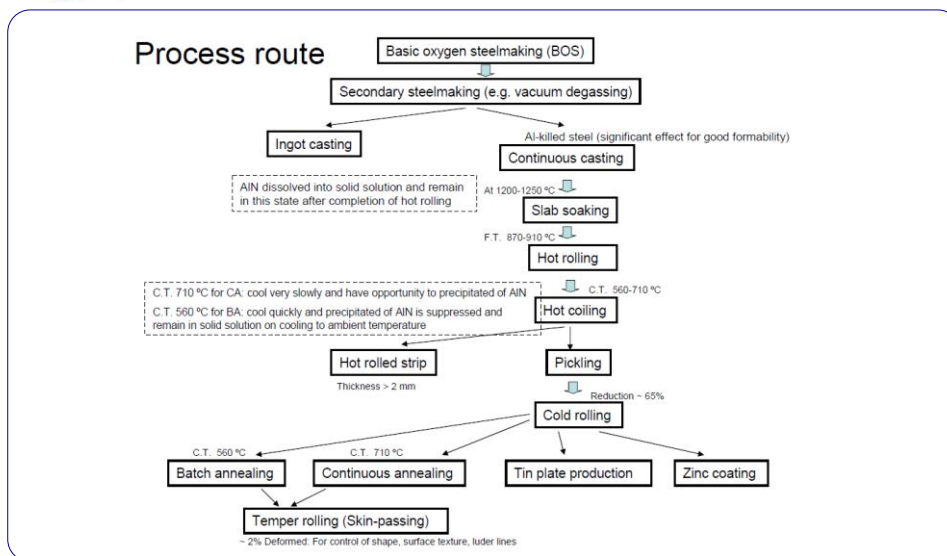
Equilibrium structure are still present but in lower amount

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Production of Low Carbon Strip Steels



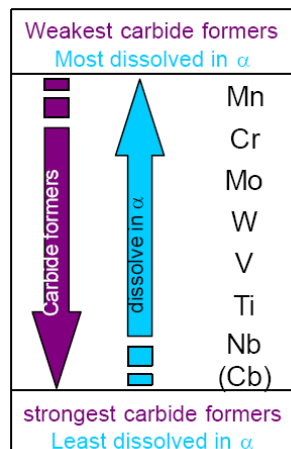
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Role of Alloy Elements in Tempered Steels

Hardenability	UTS	Toughness	Workability
Mn	Co	Mn	S
Mo	Mn	Ni	P
Cr	Mo	Mo	Mn
C	Cu	Nb	Se
B	Cr	Ta	Te
Ti	C	Zr	Pb
P	Ni	Ce	
	Nb	Cr	
	Si	Ca	
	Ta, W, V	Te, V	
	Co		



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